AI-Driven Auto-Tuning for Serverless Performance Optimization

[Cloud Computing - AI-Driven Performance Optimization]

Programme Code – Research in Computing CA1
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1. Research Problem Background

Serverless computing is innovated to transform cloud application development by providing cost-effective, auto-scaling designs that do not require infrastructure administration. Platforms such as AWS Lambda, Google Cloud Functions, and Azure Functions automatically assign resources based on demand. However, insufficient resource provisioning causes issues including cold starts, increased latency, and high execution costs (Li, et al. 2023). Traditional manual tuning techniques for optimizing memory allocation, concurrency, and execution time are not scalable (Safaryan, et al. 2022). Although Artificial Intelligence is useful and experimented with cloud computing, more advanced AI-driven techniques still should be utilized for developing machine learning system to facilitate efficiently for cloud resource allocation (Akinbolaji, 2024). Therefore, the purpose of this study is to investigate the potential effectiveness of cost and performance on enhancing serverless computing with the usefulness of AI technology.

2. Research Question

To auto-tune the serverless function performance, how can Artificial Intelligence (AI) technology be efficiently utilized, especially on optimization of resource allocation such as memory, concurrency, in terms of cost effectiveness and reducing latency of process execution? This primary question can be expanded into more specific questions:

- How can machine learning models optimize serverless configurations dynamically in real-time cloud environment?
- What are the trade-offs between AI-driven optimization, execution latency, and cloud resource costs?
- Can reinforcement learning (RL) be effectively applied to predict and adjust serverless function parameters?
- How does the tuning of AI model compare with manual configuration of traditional approach in terms of efficiency and performance?

3. Justification

The adoption of serverless computing has increased due to its ability to facilitate the performance optimization to be efficient as manual tuning is not sustainable for scalable and dynamic environments (Mongia, 2023). While leveraging machine learning models, AI provides an automated and smart solution to comprehend the complex relationships between configurations of function, workload characteristics, and performance metrics as well as the optimization of resource allocation, and accurate predictions (Syed and Albalawi, 2024). Moreover, reinforcement learning can also allow to understand the real-time adaptation for

workload variations, enhancing performance with cost-effectiveness (Schuler, et al. 2021; Mampage, et al. 2023).

This research is justified by the potential for performance improvement of serverless functions utilizing AI-driven auto-tuning which can reduce the execution latency significantly. In addition, it will provide cost saving by optimizing resource allocation to minimize unnecessary spending on compute resources. Furthermore, it will enhance scalability and elasticity as AI-powered tuning can facilitate seamless scale and adapt to fluctuating workloads for serverless applications.

4. Specific Items to be Addressed

1. Performance Bottlenecks in Serverless Computing

   Analyzing issues such as inefficient memory allocation, cold starts and high costs for process execution in well-known cloud platforms (e.g. AWS Lambda, Google Cloud Functions, Azure Functions).

2. AI-Based Predictive Optimization

   Building a machine learning model that analyses historical execution data and workload trends to predict and recommend optimal resource configurations.

3. Reinforcement Learning for Dynamic Auto-Tuning

   Creating an agent with reinforcement learning which can finetune function parameters such as memory size, timeout, concurrency in real time.

4. Experimental Evaluation

   Implementing and testing auto-tuned AI model in cloud environment utilizing the real-world workload to evaluate the effectiveness based on key metrics such as execution time, resource utilization, cost efficiency, and system stability

5. Comparison with Traditional Methods

   Evaluate and determine the advantages and limitations between AI-driven auto-tuning optimization approach and manual tuning which is performed by system administrators.

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